

ASA DataFest @ UCSD

Intro to Machine Learning

Save the Date: April 24-26, 2026

Workshop Logistics

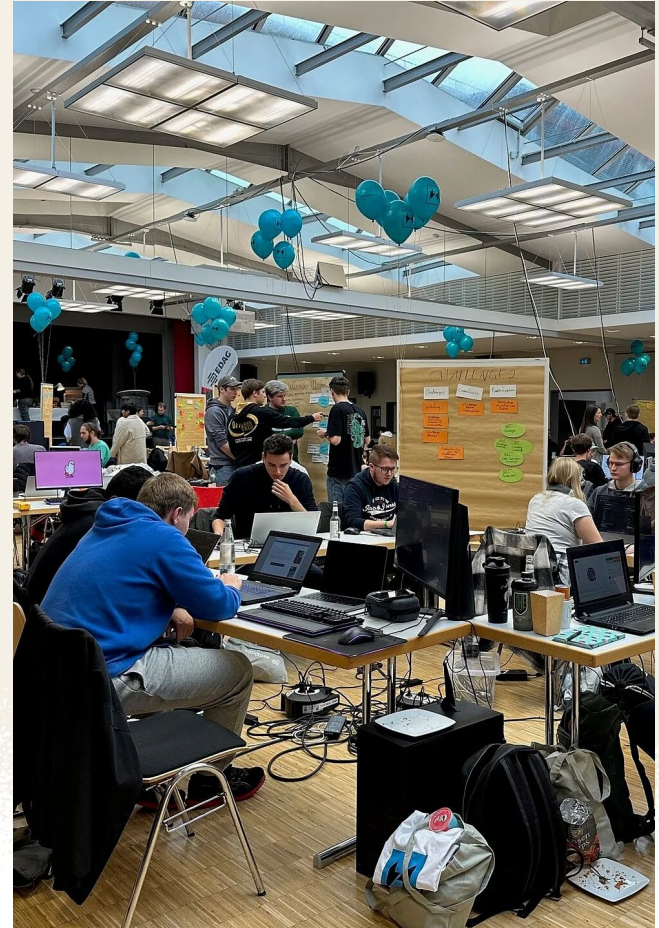
- This workshop includes three supplemental coding files (hosted in Google Colab) for students to follow along with.
- Open the following link to see each file and the associated data that will be analyzed briefly.
- Try and follow along with the slides, as each file shows real-life use cases of the methodologies talked about today!

- *Link to Notebooks/Data: [Workshop 3 Files](#)*

What is DataFest?

ASA DataFest is a multi-day, competitive hackathon for undergraduate teams. Participants work with a complex, real-world dataset, applying statistics, programming, and communication skills to uncover insights and present their findings. The event also offers networking opportunities with alumni and professionals, and awards such as Best in Show.

Founded at [UCLA](#) in 2011, it's now a national event sponsored by the American Statistical Association (ASA).



ASA DataFest at UC San Diego 2026

Datafest 2026 @ UCSD is being jointly hosted by the Department of Economics and Department of Cognitive Science, in collaboration with student organizations, including Sustainable Investment Group!

Times & Locations:

- **Day 1 (4/24):** Price Center East Ballroom
 - Check-in starting late afternoon
- **Day 2 (4/25):** Malk Hall 5th Floor
 - All day
- **Day 3 (4/26):** Malk Hall 5th Floor
 - Event ending by early afternoon

What to Expect:

- Work with a complex data set from a real-world organization.
- Mingle with graduate students, faculty, alumni and industry professionals available during the competition to offer support and guidance.
- Present your findings to a panel of judges on the final day of the competition.

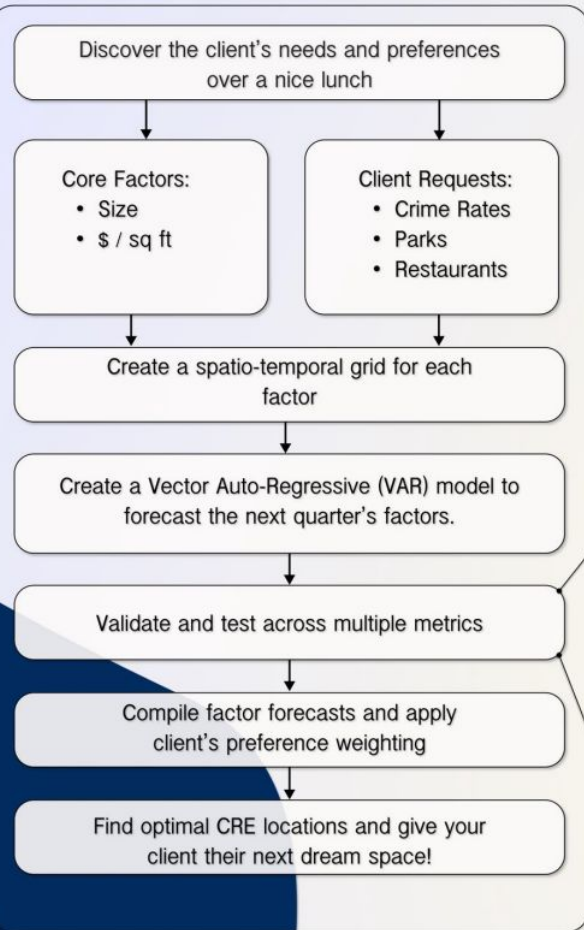
Where should x business ▼ want to put their next business?

Starbucks
REITs
Anyone

Statistically SIgnificant (B16)

Andrew Yang, Devan Velji, Sahas Goli, Steven Xu, Pranay Jha

Methodology



Case Study

A legal fintech company is seeking a large office in an up-and-coming neighborhood in Houston, Texas.

This firm prioritizes high walkability, low crime rates, and proximity to parks and restaurants.

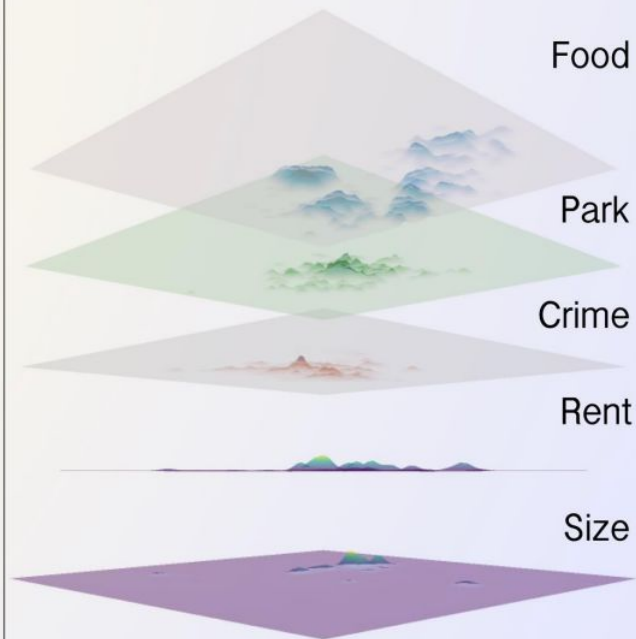
Validation

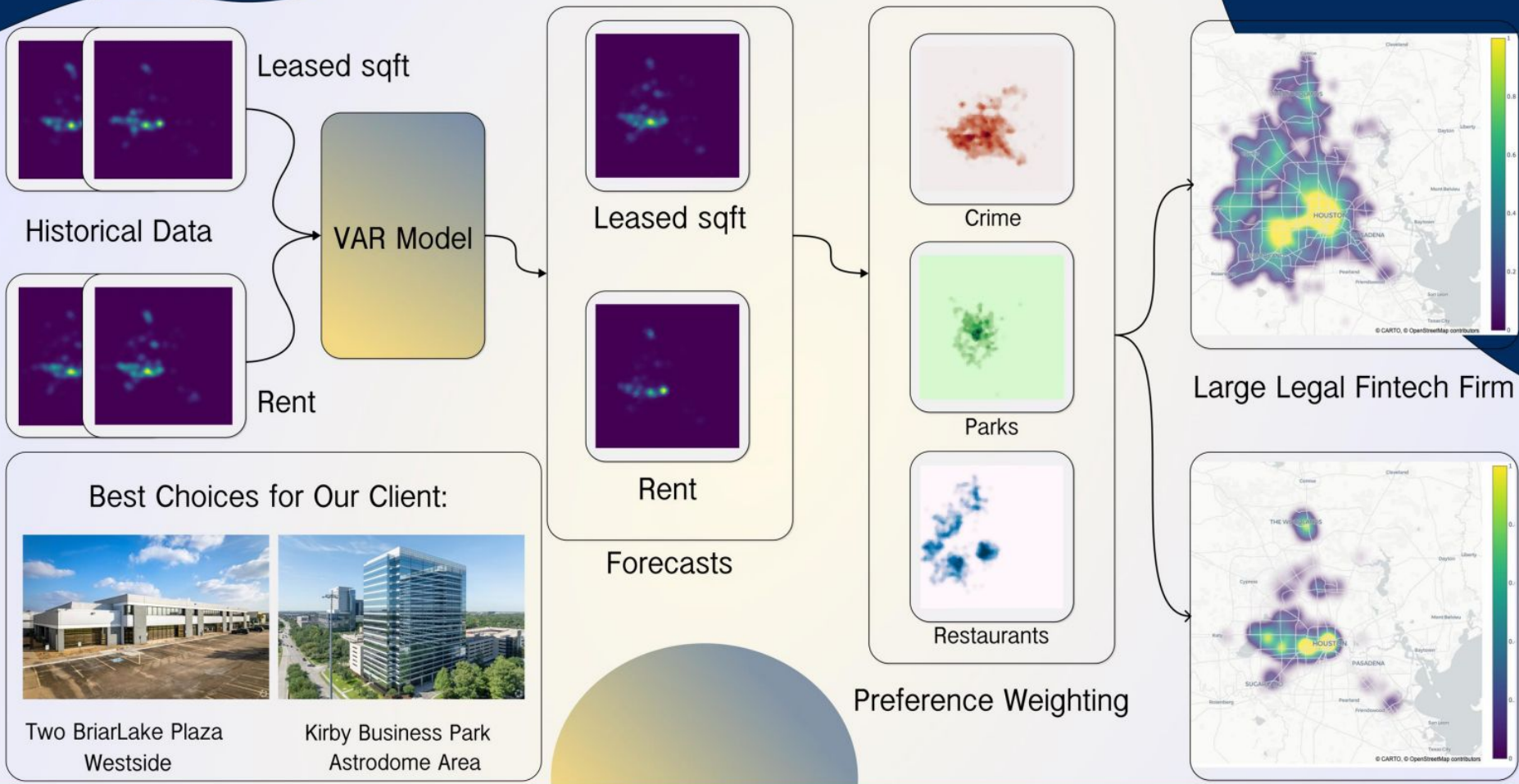
On the validation set, the VAR model performs better than our baseline model across these metrics:

- Cosine Similarity
- MSE
- Earth Mover's Distance

Factors

Layer Diagram



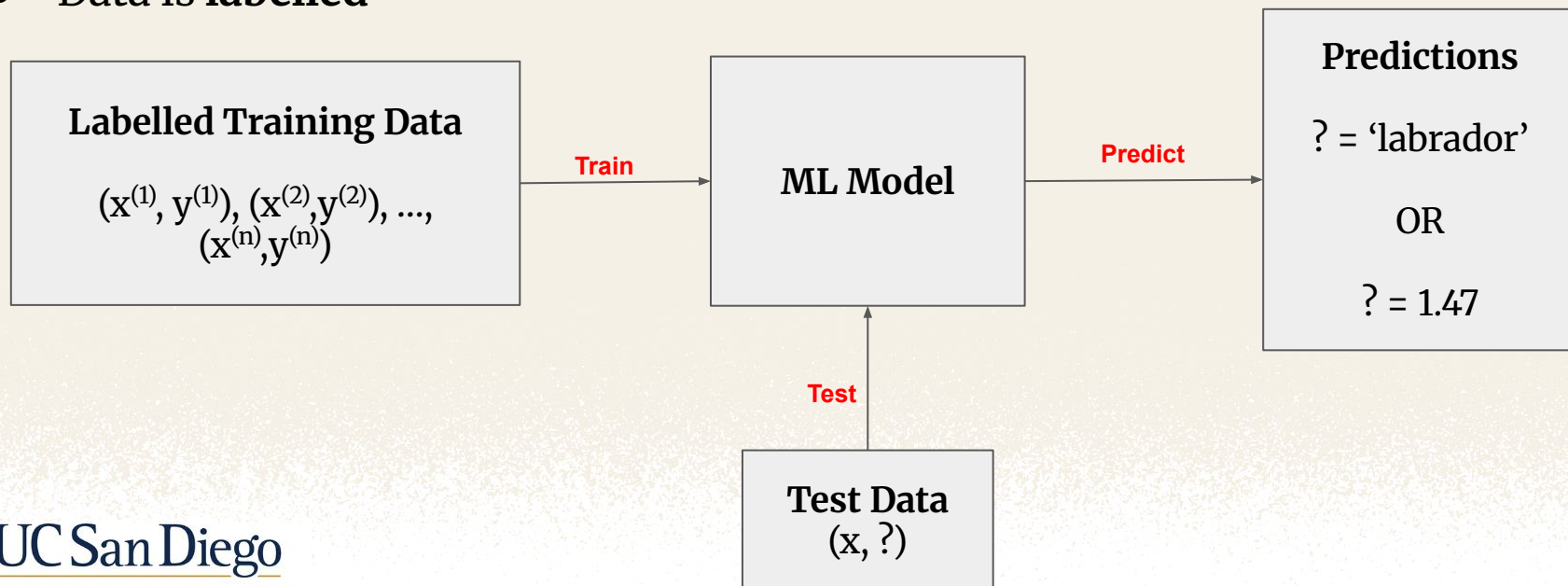


By fusing spatiotemporal forecasts with client preferences, we deliver a repeatable workflow for selecting the ideal next location based on a multitude of factors.

Arbitrary Company

Supervised learning

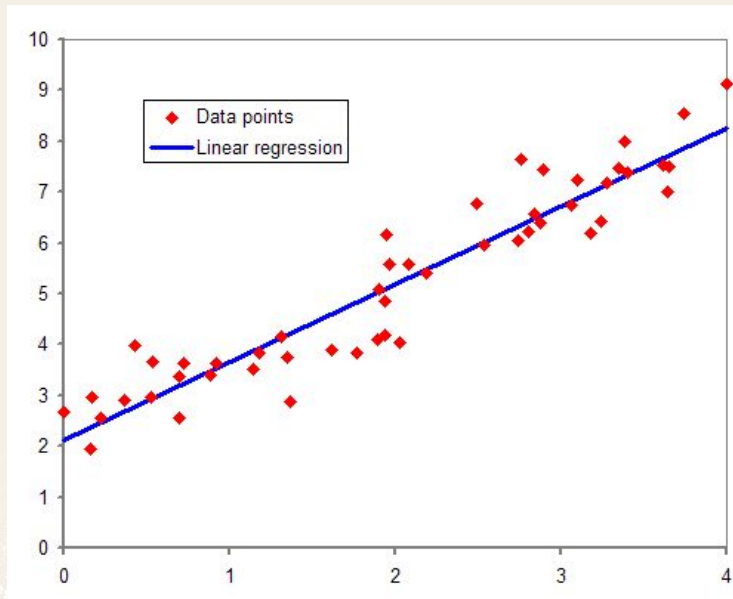
- Solving Prediction Problems
- Data is labelled



Classification vs Regression

- Classification
 - Labels (y) are discrete
 - *E.g.* spam vs not spam, type of dog breed
- Regression
 - Labels (y) are continuous
 - *E.g.* Life expectancy of person, predict AQI
- (Probability Estimation)
 - Labels (y) are probabilities
 - *E.g.* Probability email is a spam
 - How is it different to classification?

Linear Regression



- Each data point is $(x^{(i)}, y^{(i)})$
- Line formula
 - $y^{(i)} = ax^{(i)} + b$
- *Goal*: Find the line of **best fit**
- By minimising the following

$$\text{MSE}(a, b) = \frac{1}{n} \sum_{i=1}^n (y^{(i)} - (ax^{(i)} + b))^2.$$

- A closed form solution exists for the image in 2D
- But what if we want to use more features?

Multivariate Linear Regression

- Now we want to use >1 feature to predict the output
- Suppose we have 10 features (blood pressure, BMI, number of cigarettes a day etc.) and want to predict life expectancy

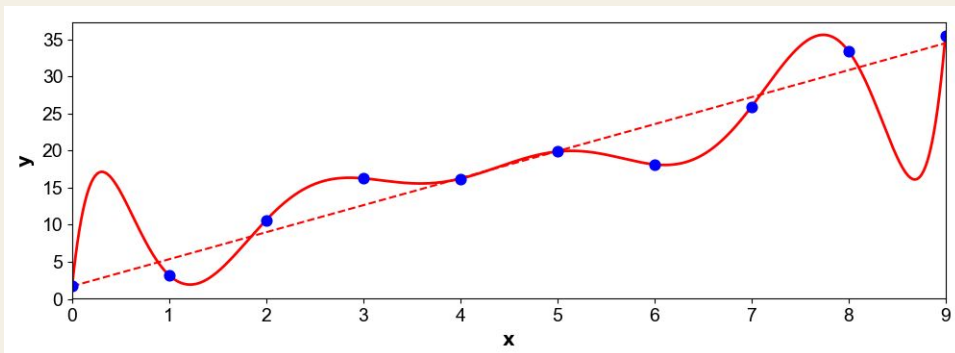
$$f(x) = w_1x_1 + w_2x_2 + \cdots + w_{10}x_{10} + b = w \cdot x + b$$

$$L(w, b) = \sum_{i=1}^n (y^{(i)} - (w \cdot x^{(i)} + b))^2.$$

But how do we minimise $L(w, b)$?

Gradient Descent: move in a direction that minimises $L(w, b)$

Regularisation



- Regularisation is a practice done to prevent ‘overfitting’.
- Model does not generalise for unseen data
- Ridge Regression

$$L(w, b) = \sum_{i=1}^n (y^{(i)} - (w \cdot x^{(i)} + b))^2 + \underbrace{\lambda \|w\|^2}_1$$

Linear Regression: Notebook

- For all practical purposes, the mechanisms of minimising $L(w,b)$ is handled by libraries like scipy or scikit-learn
- This was to introduce under what circumstances to use the regression models.

Logistic Regression

- Despite the name, this is a **classification** problem
- Specifically a binary classification problem
 - Model will help us classify if something is +ve or -ve
 - E.g. Loans: Grant or Deny (+ve or -ve)
- Different to regression (continuous value prediction vs specific class)

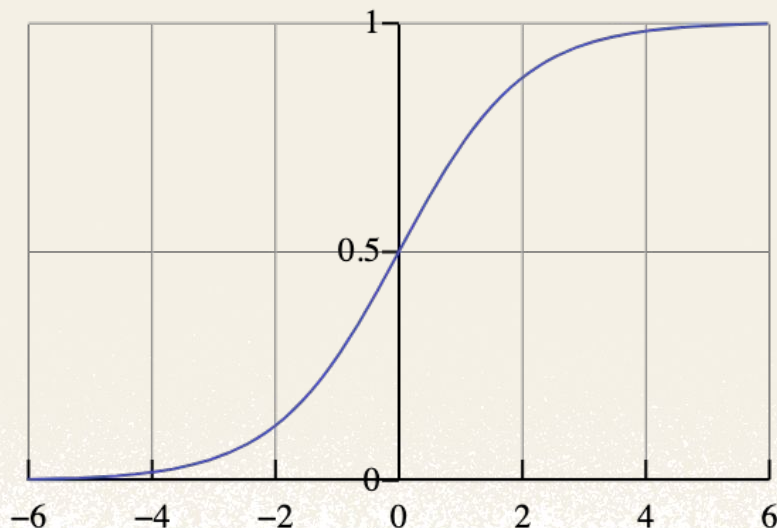
Logistic Regression

$$f(x) = w_1x_1 + w_2x_2 + \cdots + w_{10}x_{10} + b = w \cdot x + b$$

- Logistic Regression will use linear regression and tweak it to predict classes for us
- Linear Equation above (same as Lin. Reg.)
- E.g.
 - Each x is a feature like credit score, income, age, etc.
 - w and b are the parameters for the line/hyperplane
- But how do we convert the real number from $f(x)$ to a class prediction?

Logistic Regression

Sigmoid Function



- Pass $w x + b$ to the sigmoid function
- Convert value into probability
- Larger value = larger probability

$$\Pr(y = 1|x) = \frac{1}{1 + e^{-(w \cdot x + b)}}$$

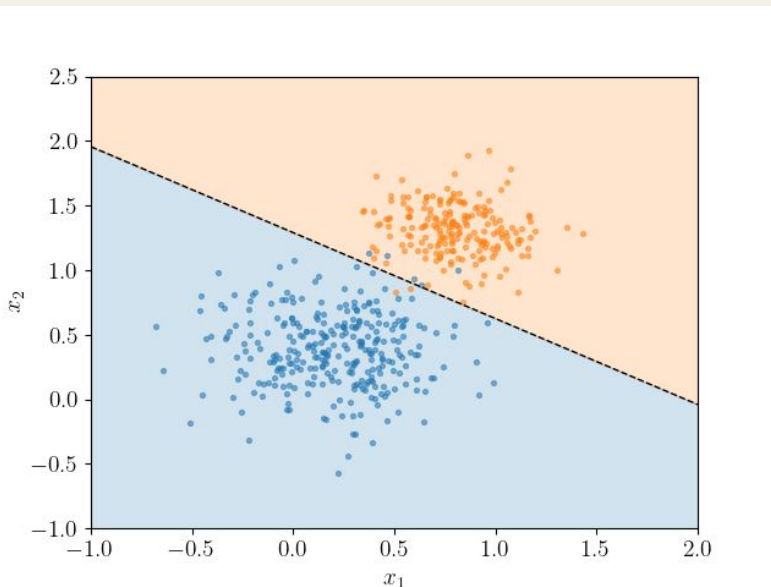
- $\{1, -1\}$ represent +ve, -ve classes
- Given the data point, what's probability of it being class 1 (+ve)?
- KEY: Model outputs probability, not class!

UC San Diego

SCHOOL OF SOCIAL SCIENCES
Department of Economics

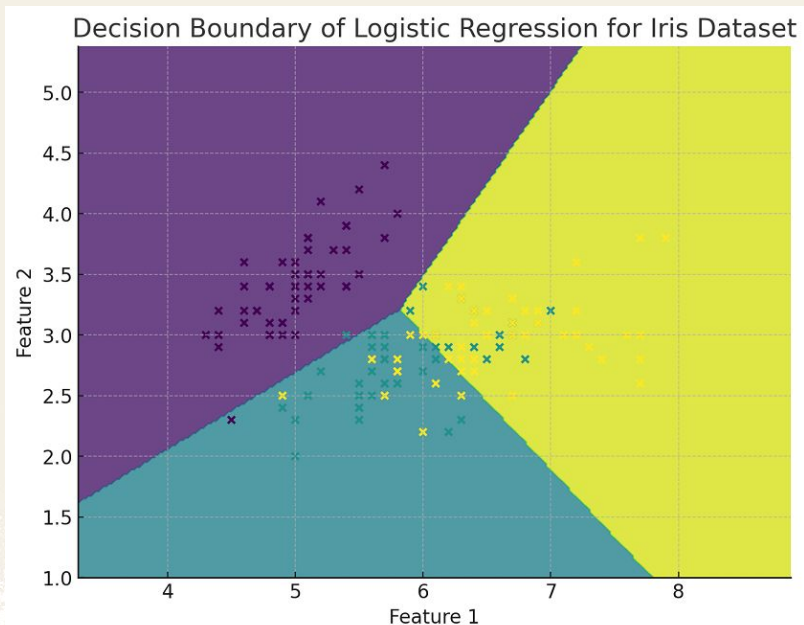
Logistic Regression

Decision Boundary



- Black line is decision boundary ($w\mathbf{x} + b = 0$)
- Represents probability = 0.5
- $w\mathbf{x} + b > 0$
 - Probability > 0.5
 - Positive Class
- $w\mathbf{x} + b < 0$
 - Probability < 0.5
 - Negative Class
- Challenge is to find $w\mathbf{x} + b = 0$ so that it separates the classes well (matches the training data)
- Not always perfect (hence probability gives confidence)

Logistic Regression



- We depicted binary classification for logistic regression
- However, you can perform **multiclass logistic regression**
- Probabilities computed using softmax:
 - Takes each raw score for each class
 - Normalises across all the classes
 - Gives you a probability for data being in that class

$$s(x_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}$$

Logistic Regression: Notebook

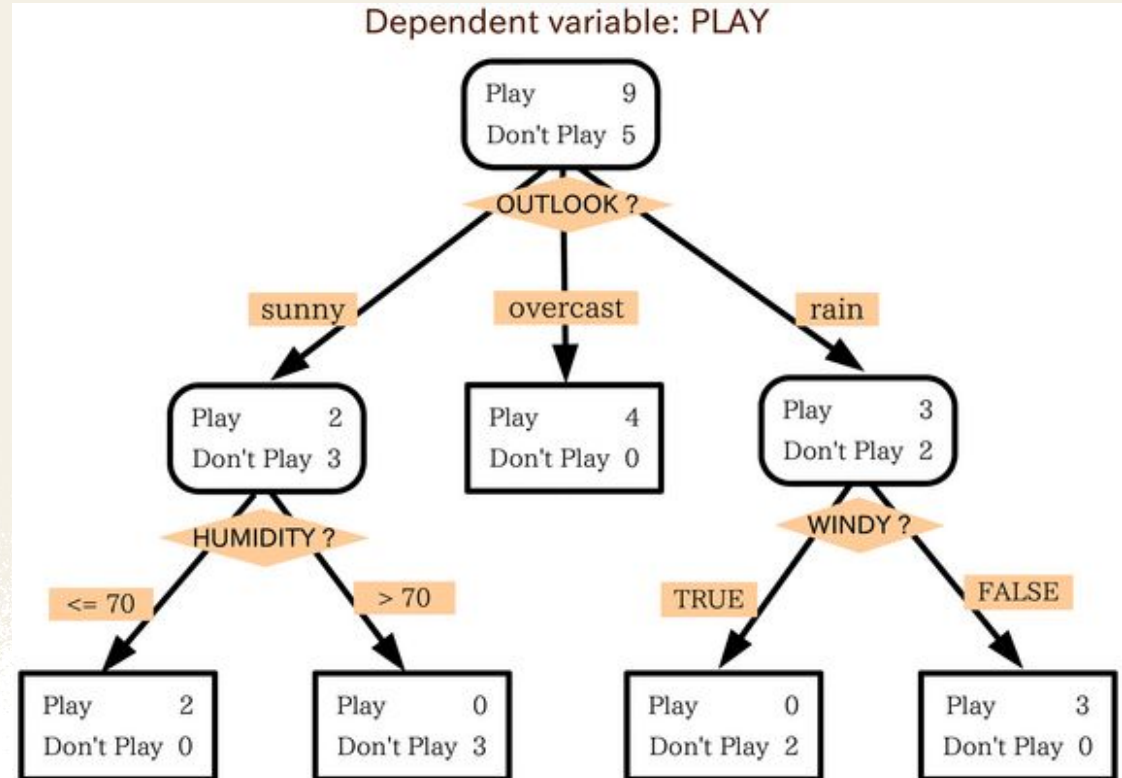
- Once again, the theory is simply there to provide you an insight on what structure of data/type of problem suits logistic regression models
- Maths and computation handled by the library but important to know the parameters and selections

Tree: Decision Tree

A decision tree is a non-parametric supervised learning method used for classification and regression. A few important concept about decision trees:

- Root node -> The single top-most node that represents the entire input space
- Internal nodes -> Each internal node represents a test on a specific feature (attribute), splits the data into subsets based on the outcome of the test
- Terminal nodes -> No outgoing edges. They represent the final output value.

Tree: Decision Tree



Tree: Decision Tree

- Let's take a peek at the Titanic data

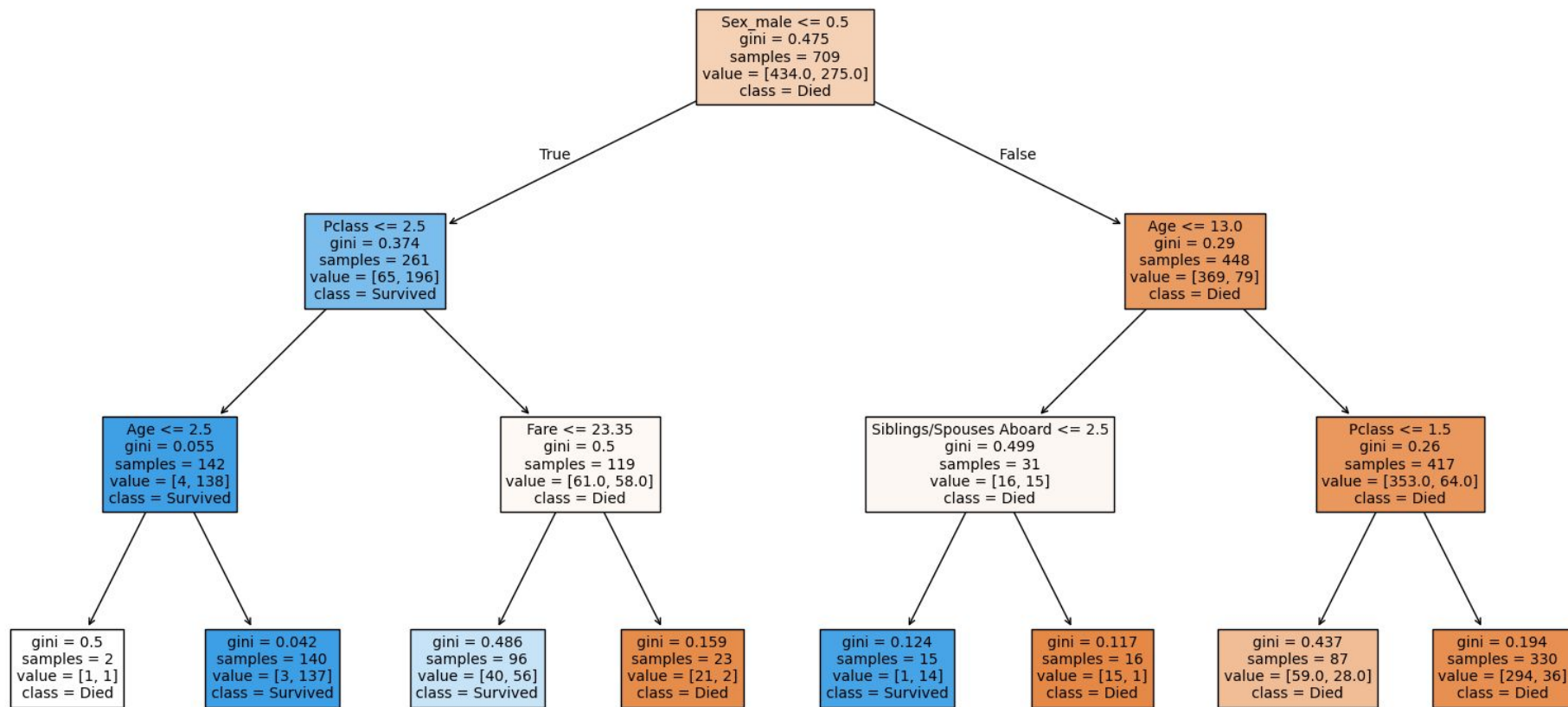
```
df = pd.read_csv('titanic.csv')
df.head(5)
```

...	Survived	Pclass	Name	Sex	Age	Siblings/Spouses Aboard	Parents/Children Aboard	Fare
0	0	3	Mr. Owen Harris Braund	male	22.0	1	0	7.2500
1	1	1	Mrs. John Bradley (Florence Briggs Thayer) Cum...	female	38.0	1	0	71.2833
2	1	3	Miss. Laina Heikkinen	female	26.0	0	0	7.9250
3	1	1	Mrs. Jacques Heath (Lily May Peel) Futrelle	female	35.0	1	0	53.1000
4	0	3	Mr. William Henry Allen	male	35.0	0	0	8.0500

UC San Diego

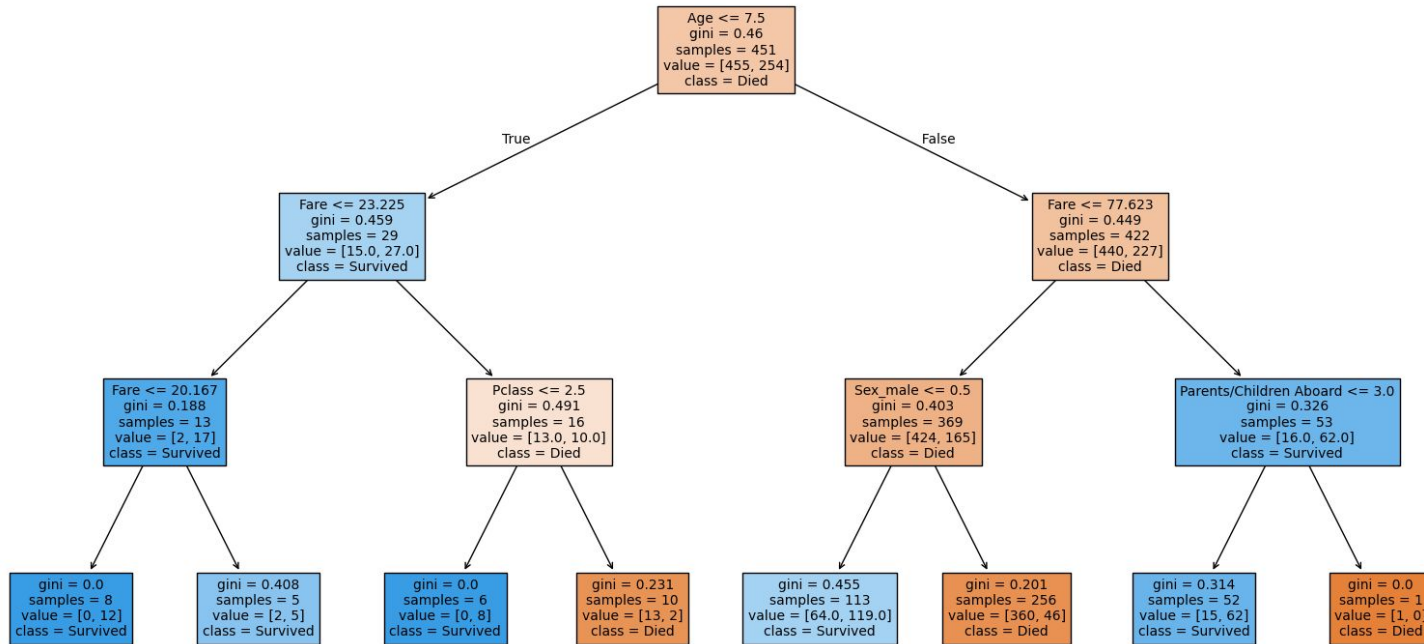
SCHOOL OF SOCIAL SCIENCES
Department of Economics

Titanic Decision Tree Logic

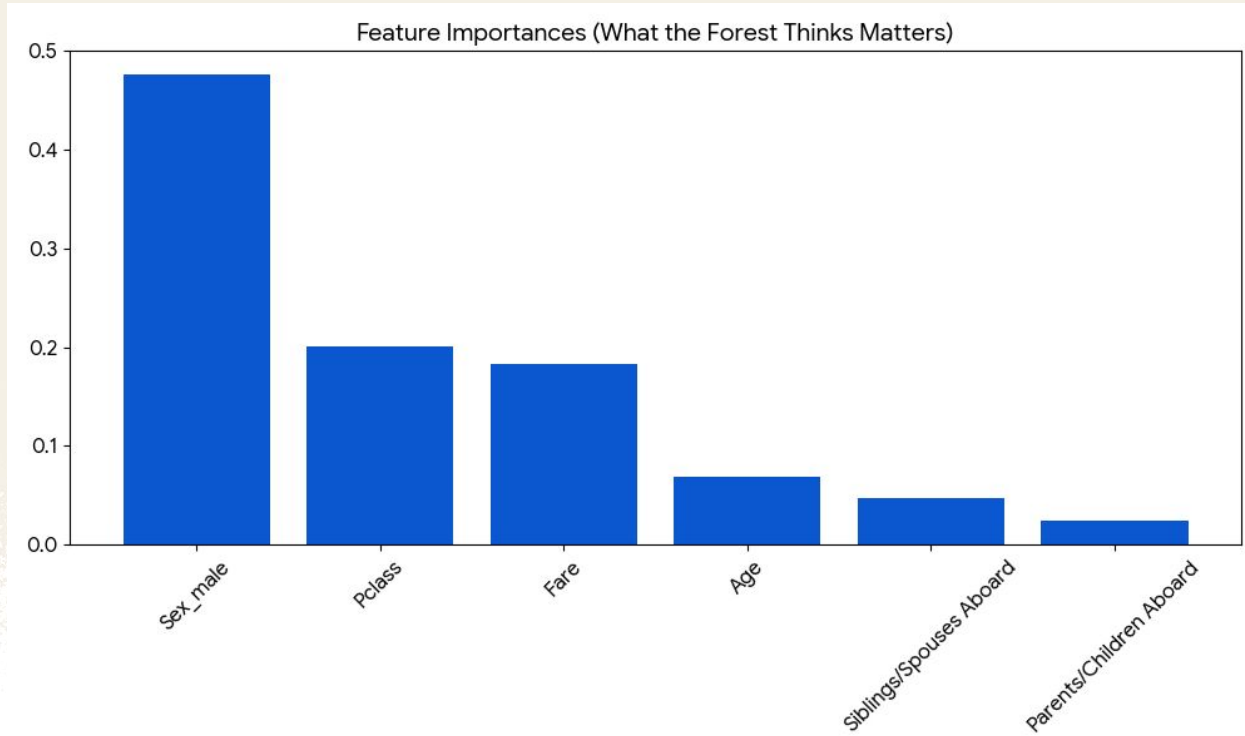


Trees: Decision Trees -> Forest

Tree #1 from the Random Forest



Forests: Feature importance



Unsupervised Learning

Unsupervised learning analyzes unlabeled data to independently discover hidden structures, patterns, or anomalies, commonly used in clustering customers or data reduction.

Common Unsupervised Learning topics includes: Principal Component Analysis (Dimensionality Reduction), K-nearest-neighbors (Clusters), RBM (Neural network)...

PCA -> KNN Pipeline

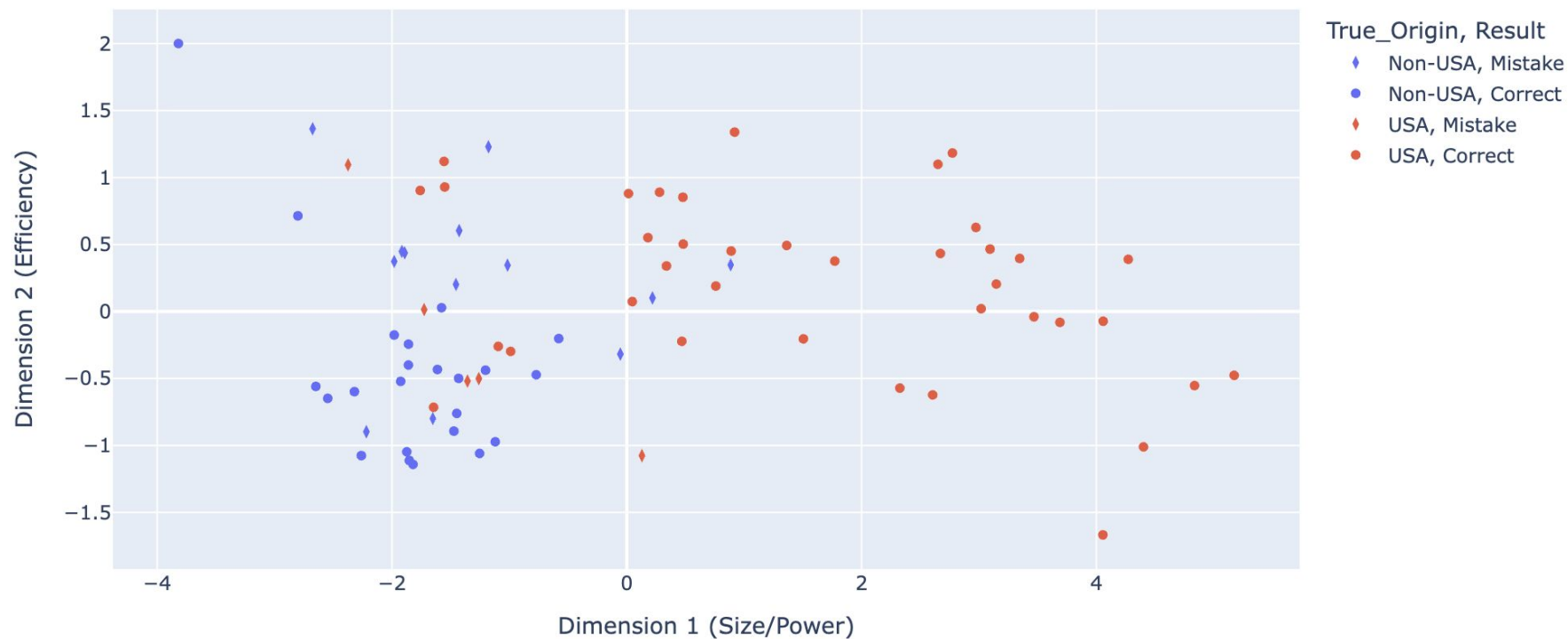
- Given a dataset with high dimension (features), we can't visualize them and find patterns
 - We can't just take out some features that we think is unimportant (we lose information)
- Then we need to have tools like PCA which help us to reduce the dimension of our data

PCA -> KNN Pipeline

```
df.head(5)
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name	origin_simple
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu	USA
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320	USA
2	18.0	8	318.0	150.0	3436	11.0	70	usa	plymouth satellite	USA
3	16.0	8	304.0	150.0	3433	12.0	70	usa	amc rebel sst	USA
4	17.0	8	302.0	140.0	3449	10.5	70	usa	ford torino	USA

PCA Map of Real Cars (Test Accuracy: 77.2%)



Frequently Asked Questions

How many people are allowed per team?

Individuals are allowed to create their own teams, and each team will have 3-5 people.

Who is allowed to participate?

This event is open to all interested undergraduate students!

Do you need statistics and coding skills to participate?

It is helpful to have taking intro courses in data science, statistics, and coding, but communication and presentation skills are also important.

Students will work in teams on 3-5, so you do not need to be skilled in everything, but you should have a team with complementary skills.

Get Involved!

Undergraduate Students:

Are you interested in participating in the DataFest competition? Fill out the student interest form to get notified when applications open for teams.

Graduate Students, Faculty, and Alumni:

Do you have experience working with complex data sets, and do you want to volunteer during the event? We'd love for you to join as a mentor! Fill out the volunteer interest form. We will follow up a month before the event to ask for your availability. We plan on setting up 2-hour shifts for mentors, and you can sign up for multiple shifts if you like!



Student Interest Form



Mentor Interest Form